

1. Conceptual Data Model

- Entities:
- Customer
- Represents an individual customer from the customer source data.
- Order
- Represents a purchase or transaction detail from the order source data.
- OrderSummary
- Represents the joined data between Customer and Order with SCD Type 2 handling.
- CustomerAggregateSpend
- Represents aggregate spend for each customer per date.
- Relationships:
- Customer to Order
- One-to-many relationship via CustId.
- Customer and Order to OrderSummary
- Joined on CustId; OrderSummary maintains SCD Type 2 history of customer attributes.
- OrderSummary to CustomerAggregateSpend
- One-to-many relationship, aggregated by Name and Date.
- Business Definitions:
- Customer: A person or entity purchasing goods/services, identified by CustId.
- Order: A transaction with itemized purchase data, linked to a customer.
- OrderSummary: Unified, historical record of customer and order details with versioning.
- CustomerAggregateSpend: Summary of total spend per customer per date.

2. Logical Data Model

- Entities and Attributes:
- Customer
- CustId (Primary Key)
- Name
- EmailId
- Region
- Order
- OrderId (Primary Key)
- ItemName
- PricePerUnit

- Qty
- Date
- CustId (Foreign Key)
- TotalAmount (Derived: $\text{PricePerUnit} \times \text{Qty}$)
- OrderSummary
- CustId
- Name
- EmailId
- Region
- OrderId
- ItemName
- PricePerUnit
- Qty
- Date
- [SCD Type 2 fields: StartDate, EndDate, Active] (implied by rules)
- CustomerAggregateSpend
- Name
- TotalAmount (Aggregated sum)
- Date
- Keys and Relationships:
- Customer.CustId = Order.CustId (1:N)
- OrderSummary is a joined, versioned entity mapping Customer and Order via CustId (1:N for Customer, 1:1 for Order per version)
- CustomerAggregateSpend groups by Name, sums TotalAmount by Date
- Cardinality and Rules:
- Each Customer may have 0 to many Orders.
- Each Order belongs to exactly 1 Customer.
- OrderSummary stores historical changes for each Customer-Order combination as per SCD Type 2.
- Each CustomerAggregateSpend record summarizes all Orders for a Customer per Date.
- Normalization Notes:
- Entities are in 3NF:
- All keys identified (Primary/Foreign, exchangeable if surrogate keys introduced)
- No partial, transitive dependencies in current structure.
- No redundant data outside history-tracking fields in OrderSummary.

- Business Rules per Entity:
- Customer: No duplicates, "Null"/Null literals removed.
- Order: No duplicates, "Null"/Null literals removed, TotalAmount calculated per row.
- OrderSummary: Update history per SCD Type 2 on customer attribute change.
- CustomerAggregateSpend: Compute sum(TotalAmount) grouped by Name and Date.

3. Physical Data Model

- Structures:
- Table: customer
 - CustId: VARCHAR/String, NOT NULL, PK
 - Name: VARCHAR/String, NOT NULL
 - EmailId: VARCHAR/String, NOT NULL
 - Region: VARCHAR/String, NOT NULL
- Table: order
 - OrderId: VARCHAR/String, NOT NULL, PK
 - ItemName: VARCHAR/String, NOT NULL
 - PricePerUnit: DECIMAL/NUMERIC, NOT NULL
 - Qty: INTEGER, NOT NULL
 - Date: DATE/TIMESTAMP, NOT NULL
 - CustId: VARCHAR/String, NOT NULL, FK → customer.CustId
 - TotalAmount: DECIMAL/NUMERIC, NOT NULL (computed: PricePerUnit × Qty)
- Table: ordersummary
 - CustId: VARCHAR/String, NOT NULL
 - Name: VARCHAR/String, NOT NULL
 - EmailId: VARCHAR/String, NOT NULL
 - Region: VARCHAR/String, NOT NULL
 - OrderId: VARCHAR/String, NOT NULL
 - ItemName: VARCHAR/String, NOT NULL
 - PricePerUnit: DECIMAL/NUMERIC, NOT NULL
 - Qty: INTEGER, NOT NULL
 - Date: DATE/TIMESTAMP, NOT NULL
 - StartDate: DATE/TIMESTAMP, NOT NULL (for SCD2, implied)
 - EndDate: DATE/TIMESTAMP, Nullable (for SCD2, implied)
 - Active: BOOLEAN, NOT NULL (for SCD2, implied)

- PK: (CustId, OrderId, StartDate)
- Table: customeraggregatespend
- Name: VARCHAR/String, NOT NULL
- TotalAmount: DECIMAL/NUMERIC, NOT NULL (AGG SUM)
- Date: DATE/TIMESTAMP, NOT NULL
- Indexing Strategy:
 - Primary keys on CustId (customer), OrderId (order), (CustId, OrderId, StartDate) for ordersummary
 - Foreign key index on order.CustId
 - Index/partition by Date and Name on customeraggregatespend for fast aggregation
- Partitioning, Clustering, Storage:
 - Partition ordersummary and customeraggregatespend by Date for optimal write/read
 - Consider clustering by CustId for customer and ordersummary
 - Delta format for all tables
 - Store computed/aggregated columns physically for efficient querying
- Constraints:
 - PK and FK as above
 - Unique constraint on OrderId in order table
 - Check constraints: Qty > 0, PricePerUnit > 0
- Performance Optimization Notes:
 - Store pre-aggregated customer spend for query acceleration
 - Use SCD Type 2 pattern for change tracking in ordersummary
 - Remove "Null"/Null values and duplicates at load time for data integrity

4. Naming Conventions

- Table and column names preserve case and terminology from source requirements
- Names are in lower_snake_case or as written in the requirements document
- No abbreviations unless stated in requirements

5. Data Dictionary

Table	Column	Data Type	Description	Nullable	Notes
customer	CustId	String	Customer unique identifier	No	PK
customer	Name	String	Customer name	No	
customer	EmailId	String	Customer email address	No	

customer	Region	String	Customer region	No	
order	OrderId	String	Order unique identifier	No	PK
order	ItemName	String	Purchased item	No	
order	PricePerUnit	Decimal	Price paid per unit	No	
order	Qty	Integer	Quantity purchased	No	
order	Date	Date	Order transaction date	No	
order	CustId	String	Customer unique identifier	No	FK → customer.CustId
order	TotalAmount	Decimal	Computed: PricePerUnit × Qty	No	
ordersummary	CustId	String	Customer unique identifier	No	Part of PK, SCD2 field
ordersummary	Name	String	Customer name	No	
ordersummary	EmailId	String	Customer email address	No	
ordersummary	Region	String	Customer region	No	
ordersummary	OrderId	String	Order unique identifier	No	
ordersummary	ItemName	String	Purchased item	No	
ordersummary	PricePerUnit	Decimal	Price per item	No	
ordersummary	Qty	Integer	Quantity	No	
ordersummary	Date	Date	Order date	No	
ordersummary	StartDate	Date	SCD2 record start date	No	SCD2 field, implied
ordersummary	EndDate	Date	SCD2 record end date	Yes	Nullable, SCD2 field, implied
ordersummary	Active	Boolean	SCD2 current record flag	No	SCD2 field, implied
customeraggregatespend	Name	String	Customer name (grouped)	No	
customeraggregatespend	TotalAmount	Decimal	Aggregated spend per customer per date	No	SUM(TotalAmount)
customeraggregatespend	Date	Date	Aggregated date	No	

6. ERD (Entity Relationship Diagram Structure)

[Customer]

*CustId

Name

EmailId

Region

[Order]

*OrderId

ItemName

PricePerUnit

Qty

Date

CustId

TotalAmount

[OrderSummary]

CustId

Name

EmailId

Region

OrderId

ItemName

PricePerUnit

Qty

Date

StartDate

EndDate

Active

[CustomerAggregateSpend]

Name

TotalAmount

Date

Customer *--1 Order

Customer *--1 OrderSummary

Order *--1 OrderSummary

OrderSummary *--* CustomerAggregateSpend

7. ERD (Text Mapping)

- Customer (1) — (0..*) Order (via CustId)
- Customer (1) — (0..*) OrderSummary (via CustId)
- Order (0..*) — (1) OrderSummary (via OrderId inclusion)
- OrderSummary (0..*) — (0..*) CustomerAggregateSpend (via Name, Date grouping)

All relationships and mappings are explicitly aligned to support historical tracking, cleansing, and aggregation as required for the enterprise data model.